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Practical - 04

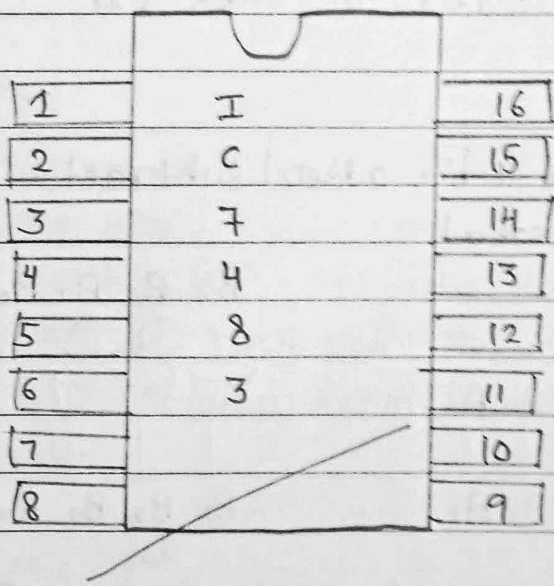
* Aim:

Realization of parallel adder/subtractor using 7483 chip.

* Theory

① Part 1

Ic 7483 is a 4-bit binary full adder which accepts two 4-bit binary A3, A2, A1, A0 and B3, B2, B1, B0 and a carry input C0 as inputs and produces a 4-bit binary sum output S3, S2, S1, S0 and a carry output C4



We can verify the functionality of the 7483 chip. Although in principle there can be 29=512 input patterns possible. We can verify all zero, all one and some other pattern.

② Part 2

Cascading of two 7483 chips to achieve addition of two 8-bit numbers $A = A_8, A_7, A_6, A_5, A_4, A_3, A_2, A_1$ and $B = B_8, B_7, B_6, B_5, B_4, B_3, B_2, B_1$ to produce $Sum = S_8, S_7, S_6, S_5, S_4, S_3, S_2, S_1$ and carry output C_9 of the 2^7 input patterns possible.

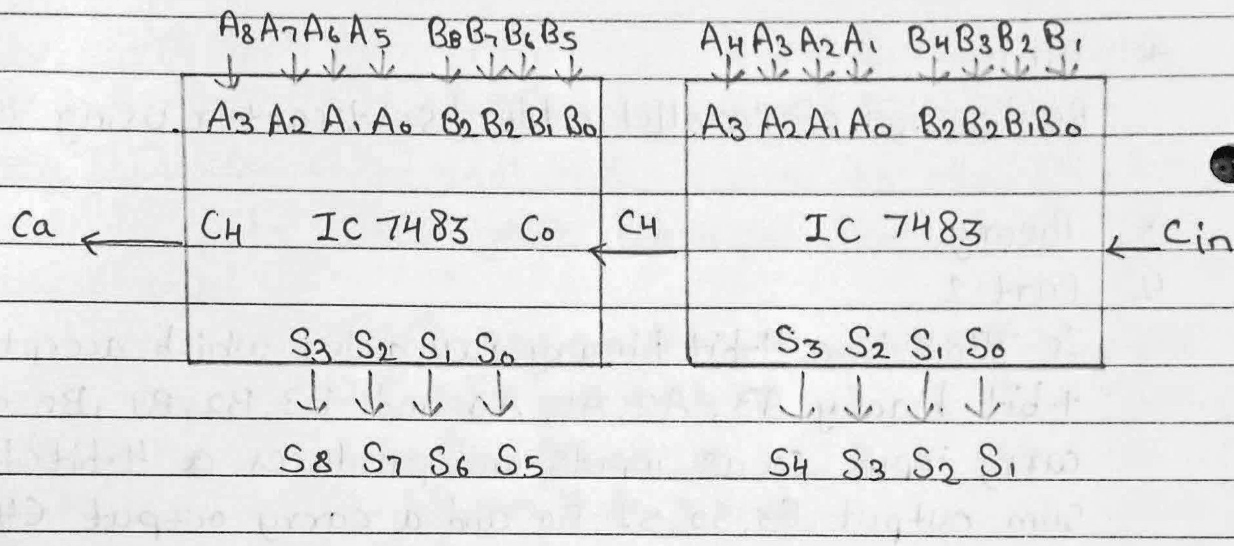


Fig: cascaded Ic 7483 (2)

③ Part 3

To implement a 4-bit adder/subtractor using 2'S complement method.

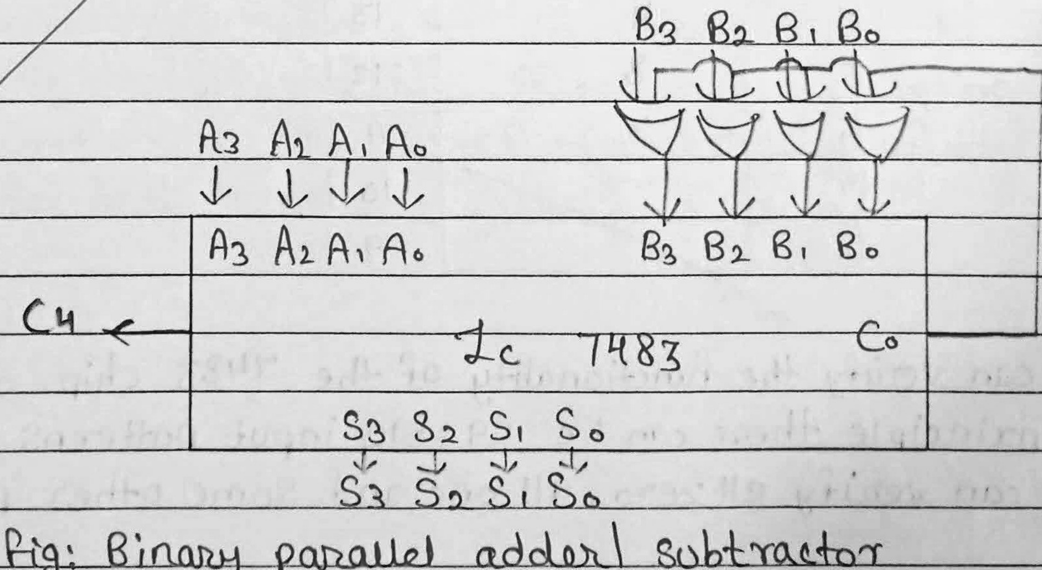


Fig: Binary parallel adder/subtractor using Ic 7483

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* Procedure

① for Part ①:

- At first click on Vcc that means Vcc = 1 and GND = 0, show message Vcc and GND properly connected.
- Next $A_0=1$, $A_1=0$, $A_2=0$, $A_3=0$, and $B_0=1$, $B_1=0$, $B_2=0$, $B_3=0$ now we see the o/p result of $S_0=0$, $S_1=1$, $S_2=0$, $S_3=0$ and $C_4=0$
- Next $A_0=0$, $A_1=0$, $A_2=0$, $A_3=0$ and $B_0=1$, $B_1=1$, $B_2=0$, $B_3=0$ now you can see the output result of $S_0=1$, $S_1=1$, $S_2=0$, $S_3=0$ and $C_4=0$
- Next, $A_0=0$, $A_1=0$, $A_2=0$, $A_3=1$ and $B_0=0$, $B_1=0$, $B_2=0$, $B_3=1$ now you can see the o/p result of $S_0=0$, $S_1=0$, $S_3=0$ and $C_4=1$
- Next $A_0=1$, $A_1=1$, $A_2=1$ and $B_0=1$, $B_1=1$, $B_2=1$, $B_3=1$ now you can see the output result of $S_0=0$, $S_1=1$, $S_2=1$, $S_3=1$

② for part ②

- At first click on the browse block button.
- next, drag the adder block and drop it onto the breadboard.
- Next, drag the bus block it onto the breadboard.
- Next, drag the o/p block and drop it onto the breadboard.
- Next make a connection from switch A_0 to A_3 of the first 1st adder.
- Next make a connection from C_4 of the 1st adder to C_0 input of the 2nd adder.
- Next make a connection from S_0 of 1st adder to 3rd LED.
- switch on the Vcc of the 1st and 2nd adder and switch on the A_0 and B_0 of 1st adder.

- Switch on the Vcc of the 1^{st} and 2^{nd} adder and Switch on the A_0, A_1, A_2, A_3 and B_0, B_1, B_2, B_3 of 1^{st} adder.
- Switch on the Vcc of the 1^{st} adder and 2^{nd} adder and Switch on A_0, A_1, A_2, A_3 and B_0, B_2, B_3 of 2^{nd} adder.
- Switch on the Vcc of 1^{st} and 2^{nd} adder and Switch on the A_0, A_1, A_2, A_3 and B_0, B_1, B_2, B_3 of 1^{st} adder.

③ for part ③
- At first click on the browse button.
- Drag the adder block and drop it onto the bread board.
- Next drag the EXOR gate block and drop it onto the bread board.
- Next drag the bus block and drop it onto the bread board.
- Next drag the o/p block and drop it onto the bread board.
- Next make a connection from Switch A to A_3 the from C_0 of adder to 2^{nd} pin of 7486.
- Next make a connection from 9^{th} pin of 7486 to 12^{th} Pin of 7486.

* Conclusion:
The experiment successfully realized parallel adder and subtractor operations. using the 7483 chip. the results matched theoretical expectations, confirming the correct functionality of the chip for both tasks.

Kulkarni